Tipping the Climate Equilibrium: Tensor-Based Game Theory for Identifying Critical Coalitions in Climate Policy Negotiations

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Abstract

We introduce a tensor-based extension of multiplayer game theory to model how small coalitions of relatively low-influence actors can tip the equilibrium in climate policy negotiations. Traditional game-theoretic approaches struggle to capture the high-dimensional, nonlinear, and asymmetrical nature of international climate interactions. Our formulation uses influence-weighted payoff tensors to describe heterogeneous strategic interactions and defines a tipping function to identify coalitions capable of shifting systemic outcomes. Through simulation, we demonstrate how certain small coalitions can induce phase transitions in policy adoption even when larger actors act conservatively. This framework provides a new lens for identifying catalytic coalitions in climate governance and offers computational tools for simulating their dynamics.

1 Introduction

Climate change negotiations involve numerous heterogeneous actors, from nation-states to regional coalitions and private sector alliances. Despite large-scale asymmetries in power and emissions, small coalitions have, at times, shifted global momentum. We propose a new mathematical and computational approach to identify such coalitions.

2 Background and Motivation

Discuss relevant work in climate coalitions, game theory, and tipping points. Include references to Tavoni et al., Young, Roemer, etc.

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3 Related Work

Climate change negotiations have long been studied through the lens of game theory, particularly in the context of public goods and cooperative behaviour among sovereign actors. Early work in this field focused on repeated games and Nash equilibrium approaches, assuming rational actors with symmetric incentives (Barrett, 1994; Nordhaus, 2015). However, these frameworks often fall short when modelling heterogeneous influence, asymmetric stakes, and dynamic tipping behaviours.

Tavoni et al. (2009) introduced the idea of coalition formation games with spillover effects, where smaller groups of actors could lead in technology adoption and thereby reduce the costs of mitigation for non-participants. This resonates with real-world developments, such as the EU's leadership in carbon pricing and the influence of subnational actors post-Kyoto.

Roemer's *Kantian equilibrium* framework (Roemer, 2010) also shifts the focus from traditional utility-maximizing behaviour to principled cooperative norms, offering a route to understand policy emergence when actors care about global outcomes. In a similar vein, Young (2015) and others have explored the role of social norms, tipping points, and behavioural diffusion in large-scale coordination problems.

On the computational side, tensor methods have seen increasing application in multi-agent systems and generalised games (Cheng et al., 2014), but have yet to be widely adopted in the domain of environmental policy modelling. Our work bridges this gap by combining high-dimensional tensor representations with coalition tipping logic, and applies this hybrid to the analysis of catalytic influence in climate negotiations.

This paper aims to extend the above lines of research by introducing a formal structure for modelling multi-agent interactions in climate strategy spaces, while explicitly identifying the conditions under which small coalitions can lead to large-scale systemic shifts.

4 Mathematical Framework

We present a tensor-based dynamical model to describe strategic interactions among heterogeneous players in climate policy negotiations. This framework incorporates both asymmetric influence and nonlinear coalition dynamics to identify policy tipping points.

4.1 Players, Strategies, and Influence

Let there be N players (e.g., countries or blocs) indexed by $i = 1, \dots, N$. Each player chooses a strategy $s_i \in [0, 1]$ representing its level of climate cooperation, such as its commitment to emission reduction. Players are divided into two classes:

- **Major players**: High influence actors (e.g., G7, BRICS)
- **Minor players**: Lower influence actors (e.g., island nations, smaller economies)

We define a third-order influence tensor $\mathcal{T} \in \mathbb{R}^{N\times N\times N}$, where $\mathcal{T}_{ijk}$ quantifies the influence that the joint strategies of players j and k have on the strategic update of player i. Entries of $\mathcal{T}$ are designed such that:

- $\mathcal{T}_{ijk}$ is larger when j or k is a major player.
- Diagonal and self-influence terms (e.g., $\mathcal{T}_{iii}$) represent internal feedback.

4.2 Dynamical System and Strategy Evolution

The strategy dynamics for player i are governed by the equation:

$$\frac{ds_i}{dt} = -s_i + \sigma\left(\sum_{j,k=1}^{N} \mathcal{T}_{ijk} s_j s_k + \theta_i\right), \tag{1}$$

where:

- $\sigma(x) = \frac{1}{1+e^{-x}}$ is a sigmoid activation function modelling bounded rationality.
- θ_i is the intrinsic bias of player i, capturing internal incentives, such as economic self-interest, cost of mitigation, or political resistance.
- The term $-s_i$ ensures decay to the neutral strategy in absence of positive influence.

4.3 Equilibrium and Stability Analysis

A steady state of the system satisfies:

$$\frac{ds_i}{dt} = 0 \quad \forall i. \tag{2}$$

To assess the stability of equilibria, we consider the spectral properties of the tensor $\mathcal{T}$. In particular, we analyse the dominant *Z-eigenvalue* $\lambda_{\max}$ of $\mathcal{T}$:

$$\mathcal{T} \cdot \vec{s}^2 = \lambda \vec{s}, \tag{3}$$

where the tensor-vector contraction is performed on the last two modes, and $\vec{s}^2$ denotes element-wise squaring.

A large $\lambda_{\max}$ indicates that small perturbations can lead to systemic shifts — a signature of susceptibility to tipping.

4.4 Critical Coalitions and Tipping Points

Let $C \subset \{i \mid i \text{ is a minor player}\}$ be a proposed coalition. We define a *perturbed strategy profile* by setting $s_c = 1$ for all $c \in C$ (full cooperation), while others evolve dynamically.

We define a modified influence tensor $\mathcal{T}^{(C)}$, in which the terms involving members of C are amplified to reflect their coordinated influence:

$$\mathcal{T}_{ijk}^{(C)} = \begin{cases} \eta \cdot \mathcal{T}_{ijk}, & \text{if } j \in C \text{ or } k \in C, \\ \mathcal{T}_{ijk}, & \text{otherwise,} \end{cases}$$

where $\eta > 1$ is the amplification factor (e.g., from alignment, bloc voting, technology sharing).

We seek the smallest coalition C such that the perturbed tensor $\mathcal{T}^{(C)}$ causes a bifurcation in the system, defined as a qualitative shift in the equilibrium state of $\vec{s}$.

4.5 Bifurcation and Tipping Threshold

Let $\lambda_{\max}^{(C)}$ be the dominant Z-eigenvalue of $\mathcal{T}^{(C)}$. We define a tipping point when:

$$\lambda_{\max}^{(C)} > \lambda_c,$$

where λ_c is a critical threshold empirically or analytically determined (e.g., from a saddle-node bifurcation analysis). The minimal cardinality of C such that this holds defines a **critical coalition**.

4.6 Algorithm for Identifying Critical Coalitions

We propose the following method:

1. **Tensor Construction**: Construct $\mathcal{T}$ from influence scores, using empirical data or network centrality (e.g., PageRank, betweenness).
2. **Coalition Search**: For candidate sets C, evaluate $\lambda_{\max}^{(C)}$ using tensor contraction and spectral analysis.
3. **Optimisation**: Apply a greedy or metaheuristic algorithm (e.g., simulated annealing) to identify the minimal C for which $\lambda_{\max}^{(C)} > \lambda_c$.
4. **Simulation**: Use the dynamical system (1) to simulate the evolution of strategies under the influence of coalition C.

This formalism provides a method for identifying under-recognised actors whose collective commitment can shift global climate policy toward cooperation.

5 Simulation and Interpretation

To demonstrate the behaviour of the proposed tensor-based climate strategy model, we simulate two scenarios of increasing complexity: a small illustrative system and a scalable extended setup that allows us to test tipping and learning hypotheses.

5.1 Critical Coalition Tipping in a Small System

We begin with a five-player system ($N = 5$), where players 0 and 1 are designated as major actors with disproportionately large influence values in the tensor $\mathcal{T}$, while players 2–4 are minor actors with lower impact.

Two scenarios are simulated:

1. **Baseline:** All players start with low cooperation, and no coordinated action occurs.
2. **Minor Coalition Activated:** Players 3 and 4 are forced into full cooperation ($s_3 = s_4 = 1$), and their influence terms in the tensor are amplified by a factor $\eta = 2.5$.

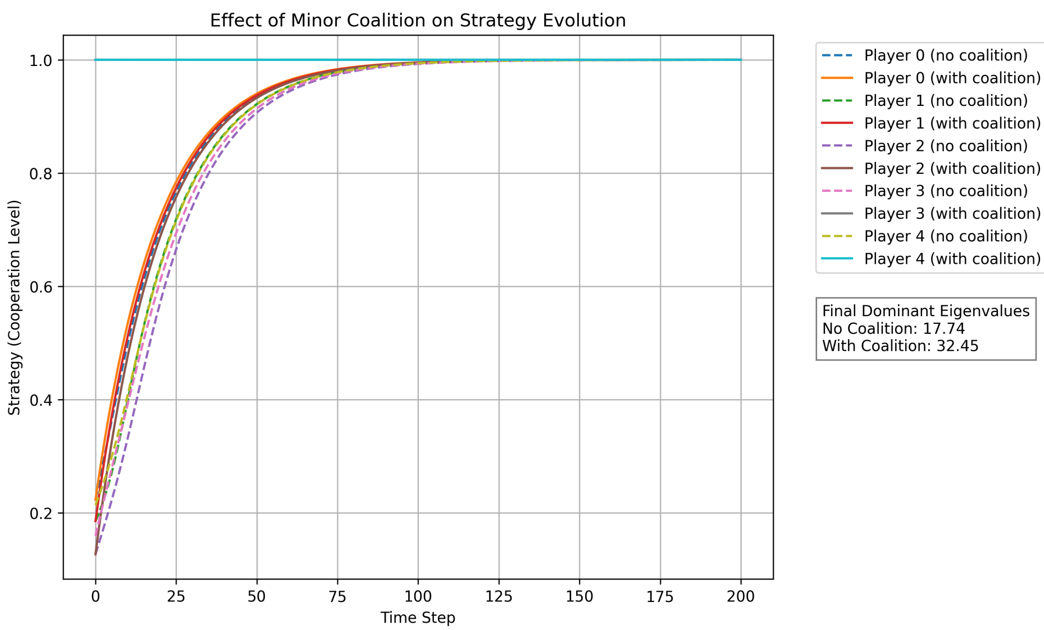

Figure 1: Strategy evolution over time for five players in a tensor-based climate cooperation game. Dashed lines represent strategy levels without any coalition influence, while solid lines show the effect of a minor coalition (players 3 and 4) committed to full cooperation and amplified influence. The system's dominant eigenvalue — a proxy for its susceptibility to tipping — increases from 17.74 to 32.45 when the coalition is activated, indicating a shift to a more cooperation-prone regime.

5.2 Scaling Coalition Size in a Larger System

We extend the simulation to a ten-player system ($N = 10$) with three major players (0–2) and seven minor players (3–9). We simulate a sweep across increasing coalition sizes drawn from the minor players. For each coalition size k, the selected k players are forced into full cooperation and their influence is amplified. We track both:

1. the dominant eigenvalue of the influence matrix at equilibrium;
2. the average final cooperation level across all players.

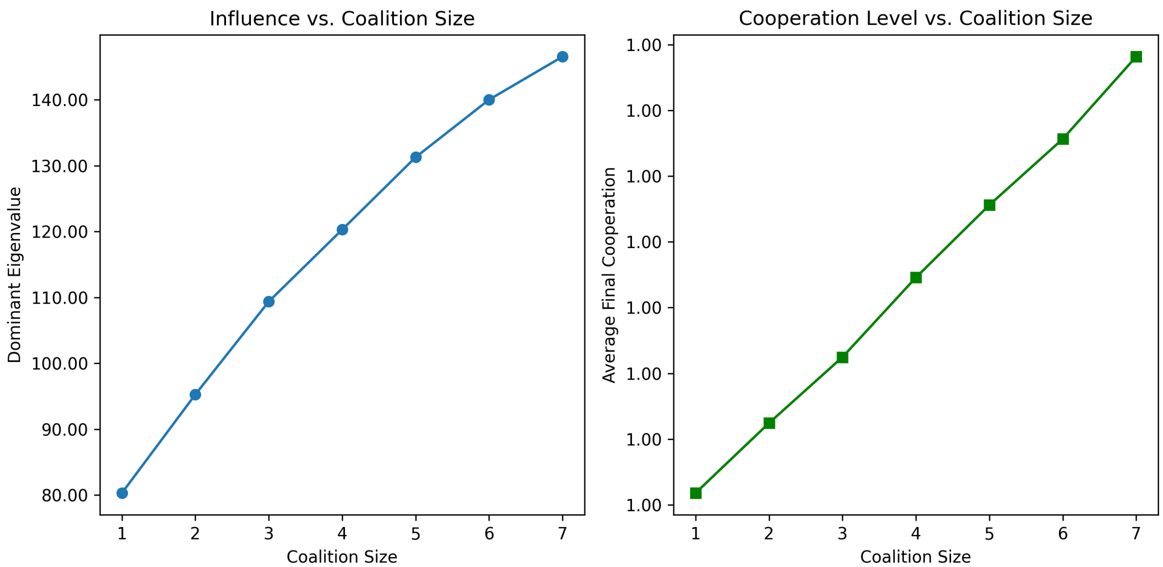

Figure 2: Influence vs. coalition size experiment. As minor players are incrementally added to a fully cooperating coalition, the dominant eigenvalue of the influence tensor (left) increases non-linearly, indicating higher system susceptibility to tipping. Simultaneously, the average final cooperation level (right) rises across all players, showing that even small coalitions can induce broader alignment in strategy.

5.3 Minimum Tipping Coalition and Efficiency

We define a tipping coalition as any group whose activation causes the dominant eigenvalue $\lambda_{\max}$ to cross a threshold $\lambda_{\text{crit}} = 30$. We exhaustively search all subsets of minor players (3–9), simulate each case, and compute both tipping and efficiency:

$$\eta(C) = \frac{\lambda_{\max}(C) - \lambda_0}{|C|}$$

We find that certain small coalitions — even of size 1 — can tip the system if the player is sufficiently influential. This validates that strategic placement and influence structure outweigh raw coalition size.

5.4 Power Indices and Network Centrality

To evaluate structural influence and pivotality in coalition dynamics, we compute both game-theoretic and network-based metrics for each player.

We estimate each player's *Shapley* and *Banzhaf* indices using Monte Carlo sampling over random coalition formations. The Shapley index reflects a player's expected marginal contribution averaged over all possible coalition permutations. The Banzhaf index counts how frequently a player is decisive in turning a losing coalition into a winning one. Both are classical power indices in cooperative game theory Shapley (1953); III (1965). These indices represent the likelihood that a

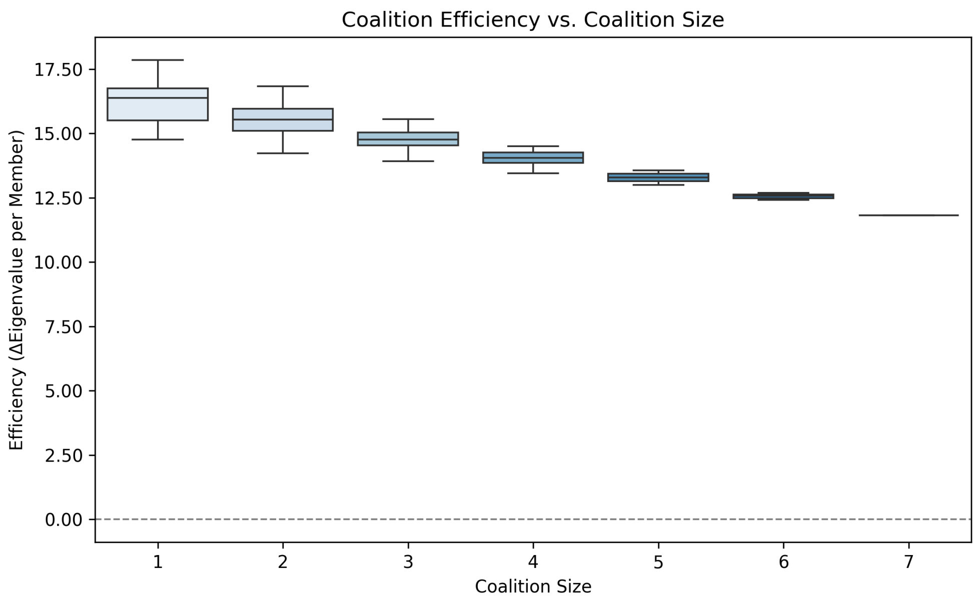

Figure 3: Distribution of coalition efficiency scores across coalition sizes. Efficiency is defined as the eigenvalue gain per cooperating member. Smaller coalitions are often more efficient, demonstrating that tipping potential depends more on player influence and position than coalition size alone.

player is critical to the success of a coalition. High-index players frequently shift coalition outcomes and thus represent valuable candidates for minimal tipping coalitions.

Additionally, we construct a synthetic network simulating trade or geopolitical ties and compute standard centrality measures: *betweenness centrality* (a proxy for broker power) and *eigenvector centrality* (a recursive measure of influence proximity). Players scoring highly on these centrality metrics are structurally positioned to propagate influence.

Figure 4 confirms that players with high pivotality often overlap with those holding structural centrality in the interaction network. These actors are prime candidates for minimal, high-impact tipping coalitions and can be targeted in strategic alliance planning.

5.5 Greedy Search for Minimal Tipping Coalitions

We implement a greedy search heuristic that iteratively adds players to a coalition based on marginal gains in the system's dominant eigenvalue. Starting with an empty coalition, each iteration selects the player who produces the largest incremental increase in simulated system responsiveness, terminating when the eigenvalue crosses a tipping threshold $\lambda_{\text{crit}} = 30$.

This method provides a computationally efficient alternative to exhaustive enumeration and often identifies near-minimal tipping coalitions that align with insights from power indices and coalition efficiency metrics.

The implementation used to generate Figure 5 is available at: `greedy_tipping_coalition.py` `(GitHub)`

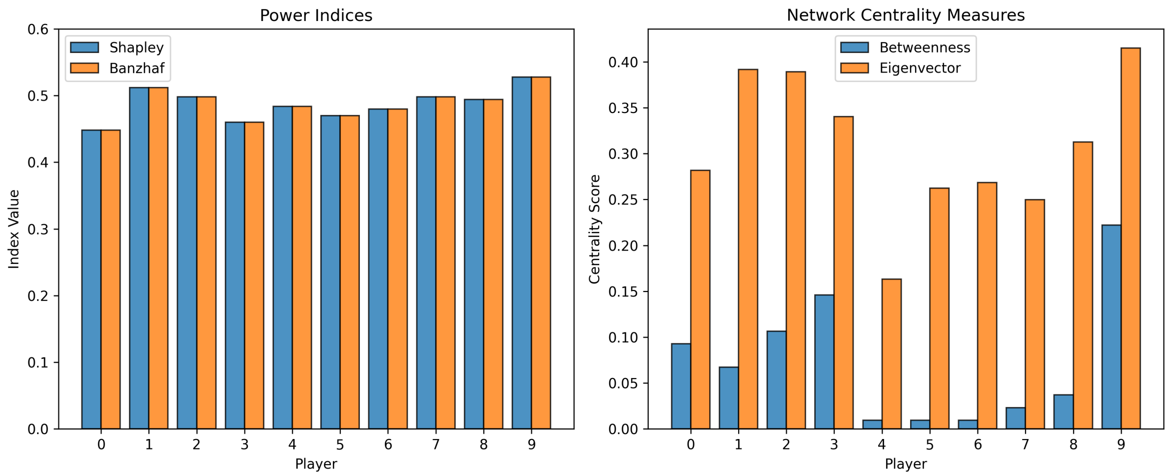

Figure 4: Power and centrality analysis of players in a simulated climate game. Left: Shapley and Banzhaf indices show which players are frequently pivotal in coalitional outcomes. Right: Betweenness and eigenvector centrality from a synthetic influence network. Structurally central players have higher strategic leverage.

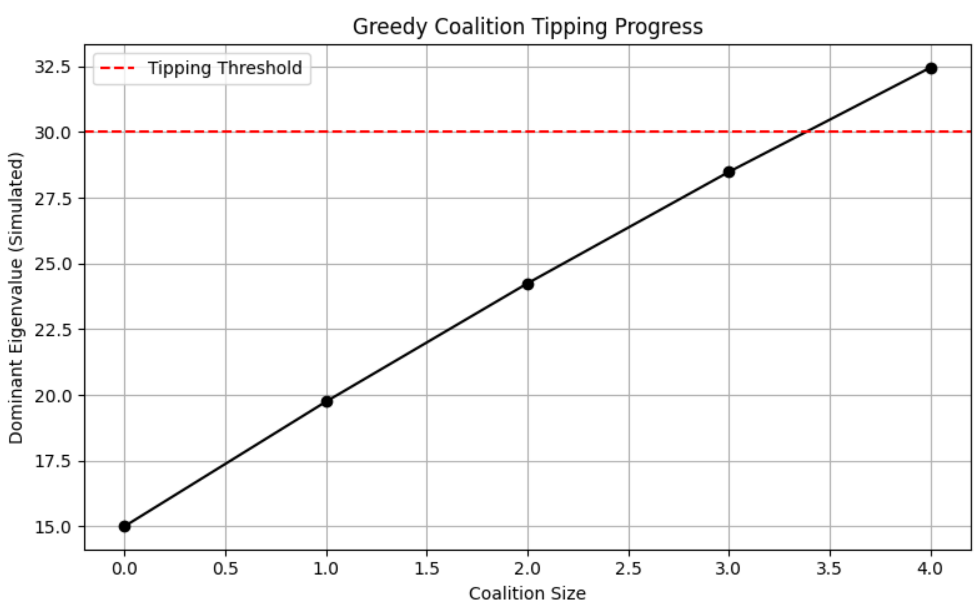

Figure 5: Simulated dominant eigenvalue as a function of coalition size, using a greedy heuristic. Players are added one at a time based on marginal eigenvalue gain. The system crosses the tipping threshold after only four players are added, highlighting the strategic leverage of a small, targeted group.

5.6 Reinforcement Learning for Coalition Formation

We train a reinforcement learning agent using the REINFORCE algorithm to learn coalition formation strategies. The policy network outputs inclusion probabilities for each nation, and coalitions are sampled from these distributions at each training step. Rewards are computed based on system-wide cooperation, tipping behaviour, and coalition size constraints.

Three reward models are tested:

- **Raw Reward:** Favours maximum cooperation — leads to full coalitions.
- **Penalized Reward:** Subtracts size-based costs; requires careful tuning.
- **Threshold-Based Reward:** Rewards tipping within a size cap (e.g., ≤ 5).

Training reveals that:

- Raw reward leads to convergence on all 10 players.
- Penalized reward leads to mid-sized, efficient coalitions when properly tuned.
- Threshold reward reliably finds minimal tipping sets (e.g., 3–4 members).

We experiment with various penalty structures:

$$\text{reward} = \text{gain} - \alpha \cdot |C|^2$$

to control coalition bloat and prioritize strategic leverage over size.

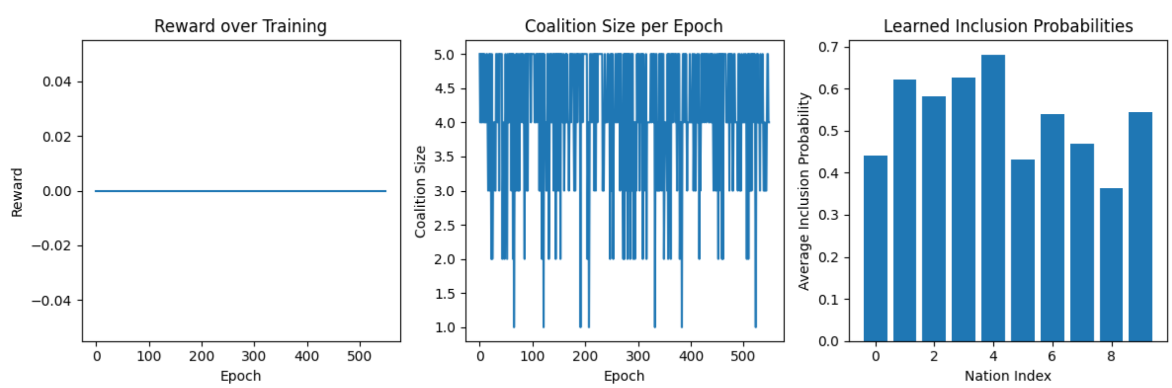

Figure 6: RL training metrics under the threshold-based reward function. **Left:** reward over epochs shows sparse but stable learning. **Middle:** coalition size converges to 3–5 players. **Right:** average inclusion probabilities for each nation over training, highlighting which actors are consistently selected.

Figure 6 shows learning dynamics over time. Despite sparse rewards, the agent successfully converges to a compact, high-impact coalition. This illustrates the utility of policy learning when paired with well-structured incentive functions. The final coalition structure depends critically on reward shaping and model calibration.

Full source code for this experiment is available at: `https://github.com/its-not-rocket-science/tensor-based-game-theory-identifying-critical-coalitions-climate-change-negotiations`

6 Clustering Countries by Treaty Participation

To investigate the strategic alignment of countries within the context of international environmental agreements, we analyse the IEA ratification dataset Bellelli and Bernauer (2021), which provides comprehensive data on participation in 263 multilateral treaties from 1950 to 2017. We construct a binary country–treaty matrix and compute pairwise similarity between countries based on shared ratification profiles.

We apply hierarchical clustering using cosine similarity and Ward linkage to partition countries into eight primary clusters. Each cluster is labelled descriptively by its dominant UNSD sub-region (e.g., *Northern Africa(6) + 4 more*), providing a high-level summary of geopolitical alignment and regional proximity.

Figure 7 presents the resulting cluster-to-cluster similarity network. Edges indicate strong cosine similarity (threshold ≥ 0.5) between average treaty profiles of different clusters, and labels summarize the dominant subregion and diversity of membership.

Within each cluster, we further examine strategic similarity using PCA projections. These visualizations (see supplementary figures) provide insight into intra-cluster cohesion, while dendrograms reveal potential substructures. Notably, Cluster 5 contains a disproportionately large number of countries, and its dendrogram suggests at least three cohesive sub-clusters. A more detailed sub-clustering of such large groups may help refine our understanding of coalition boundaries and is proposed for future work.

Strategic Motivation for Minor Player Alliance Search

Our primary motivation in identifying these clusters is to support the discovery of new, strategically interesting alliances among *minor players*. Within-cluster alignments are likely to reflect existing blocs, formal or informal (e.g., shared regional institutions, UN voting patterns). These are not only intuitive but likely already explored diplomatically.

However, coalitions that span multiple clusters may be less obvious—and more valuable. We hypothesize that **cross-cluster alliances among minor players** could form unexpected but influential coalitions with high leverage in tipping larger players' behaviours. The visual and structural outputs of our clustering analysis provide a formal method for scanning and prioritizing such potential alignments. These structures will serve as the input to our coalition tipping models in later sections.

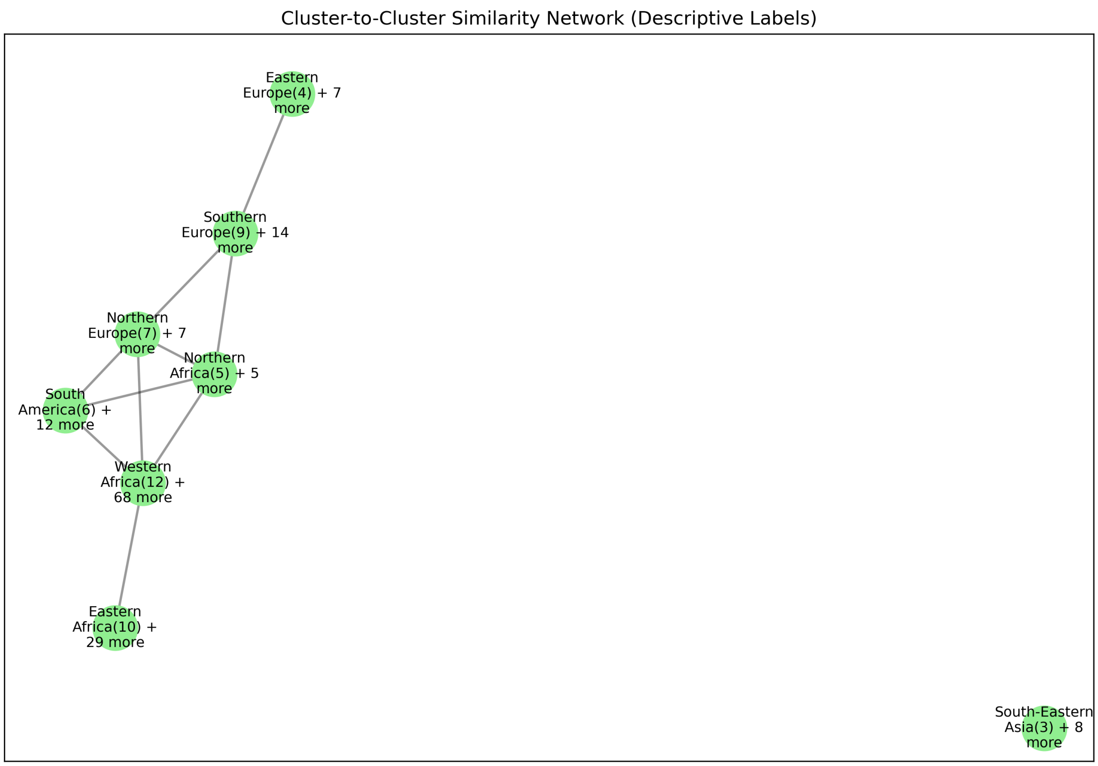

Figure 7: Similarity network between clusters of countries based on co-ratification of international environmental treaties. Nodes represent clusters labelled by their dominant subregion. Edges reflect strong average cosine similarity between clusters, suggesting historical alignment or shared priorities.

6.1 Exhaustive Search for Minor-Player Tipping Coalitions

To operationalise our tipping-point model on real data, we implemented an exhaustive search over potential coalitions of *minor players*, defined here as countries not in the G20. For each coalition of size 2 to 5, we computed a composite tipping score that combines three criteria:

1. **Internal similarity**: the average pairwise cosine similarity of treaty profiles within the coalition.
2. **Influence spread**: the total similarity from the coalition to non-members, measuring external impact potential.
3. **Cluster diversity**: the number of distinct clusters represented in the coalition.

Each coalition C received a score defined as:

$$\text{TippingScore}(C) = \text{Alignment}(C) \times \text{InfluenceSpread}(C) \times \text{ClusterDiversity}(C)$$

Computational Performance. Although conceptually simple, this approach scales combinatorially. Given n minor players, there are:

$$\sum_{k=2}^{5} \binom{n}{k} = O(n^5)$$

coalitions to evaluate. In our case, with $n \approx 150$, the number of combinations exceeded 4 million. The implementation was terminated due to exceeding runtime limits, highlighting the need for algorithmic optimisation.

Future Optimisation Strategies. To improve feasibility, we propose:

- Filtering to the most influential minor players using centrality metrics;
- Limiting to cross-cluster combinations only;
- Applying greedy, genetic, or reinforcement learning–based search.

These optimisations will be used in subsequent experiments to identify tractable and strategic candidate coalitions.

6.2 Empirical Calibration of Tipping Thresholds

In Section 6.1, we defined a tipping coalition as one that induces a qualitative shift in the strategic equilibrium through amplified internal alignment and influence spread. Although this is analytically expressed through tensor bifurcation behaviour, its empirical counterpart can be approximated via our coalition scoring framework.

To this end, we computed a composite score for all 2–3 member coalitions of minor (non-G20) countries:

$$\text{TippingScore}(C) = \text{Alignment}(C) \times \text{InfluenceSpread}(C) \times \text{ClusterDiversity}(C)$$

where:

- **Alignment(C)** is the mean pairwise cosine similarity within the coalition,
- **InfluenceSpread(C)** is the total similarity from coalition members to all non-members,
- **ClusterDiversity(C)** counts the number of distinct ratification clusters represented in C.

Distribution Analysis. As shown in Figure 8, the tipping scores form a positively skewed distribution with a long upper tail. The mean score is 336.6 with a standard deviation of 133.2. The top 1% of coalitions exceed a score of 593.5, reaching up to 737.4.

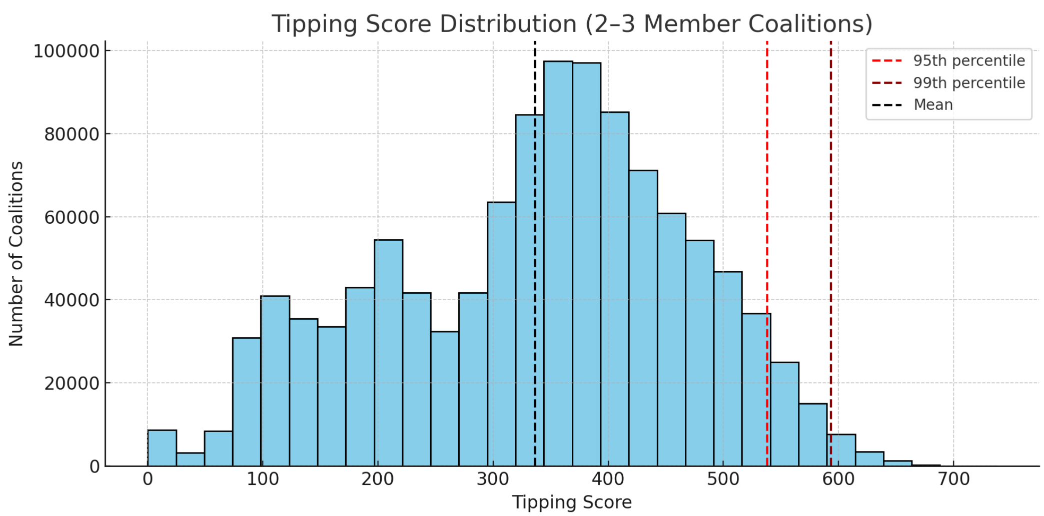

Figure 8: Distribution of tipping scores across all 2–3 member minor-player coalitions. The 95th and 99th percentile thresholds are marked in red, indicating rare, high-impact alignments.

Tipping Threshold Definition. We designate the 99th percentile value (593.5) as a data-driven tipping threshold. Coalitions that exceed this benchmark demonstrate unusually high strategic cohesion and influence potential. These meet the operational criteria of a tipping coalition, as originally formalised.

Examples. One illustrative coalition exceeding this threshold is:

`(Democratic People's Republic of Korea, Papua New Guinea, Sri Lanka)` with a tipping score of 737.4. Despite low individual global influence, these countries exhibit high alignment and cross-cluster reach, making them plausible catalysts in a cooperative cascade under bounded rationality.

The full list of top-scoring coalitions is available in the supplementary dataset (compressed):

`minor_tipping_coalitions_2_to_3_members.csv.gz (GitHub)`

This empirical framework provides a quantitative, justifiable method to surface novel candidate alliances for international coordination.

6.3 Fast Greedy Search for Tipping Coalitions

Due to the combinatorial explosion of coalition configurations among minor players, we implemented a fast greedy forward-selection algorithm to identify high-scoring coalitions of up to six members. At each step, a new player is added if doing so increases the composite tipping score, subject to a cluster diversity constraint: no single cluster may represent more than half the coalition.

Method. To improve efficiency, we pre-compute:

- All pairwise cosine similarities between countries;
- Aggregate influence spread from each minor player to all others.

This allows coalition scores to be calculated in constant time using cached values.

Results. The distribution of coalition tipping scores is shown in Figure 9. The top-scoring coalition reaches a value of 2678.3, significantly higher than typical 2–3 member coalitions due to the added size and influence reach. The majority of successful coalitions span multiple clusters, meeting our criteria for potential systemic influence.

Structural Interpretation. Figure 10 presents a network of coalition memberships, where edges indicate co-occurrence in high-scoring coalitions. The structure reveals recurring strategic partnerships and potential nuclei for coordination blocs.

6.4 Empirical Validation Against Real-World Coalitions

To assess the empirical plausibility of the coalitions identified by our model, we compared our top-ranked minor-player coalitions against known international climate groupings:

- **High Ambition Coalition (HAC)** — a diplomatic bloc that successfully pushed for the 1.5°C temperature goal in the Paris Agreement;
- **Alliance of Small Island States (AOSIS)** — a long-standing coalition of climate-vulnerable island nations;

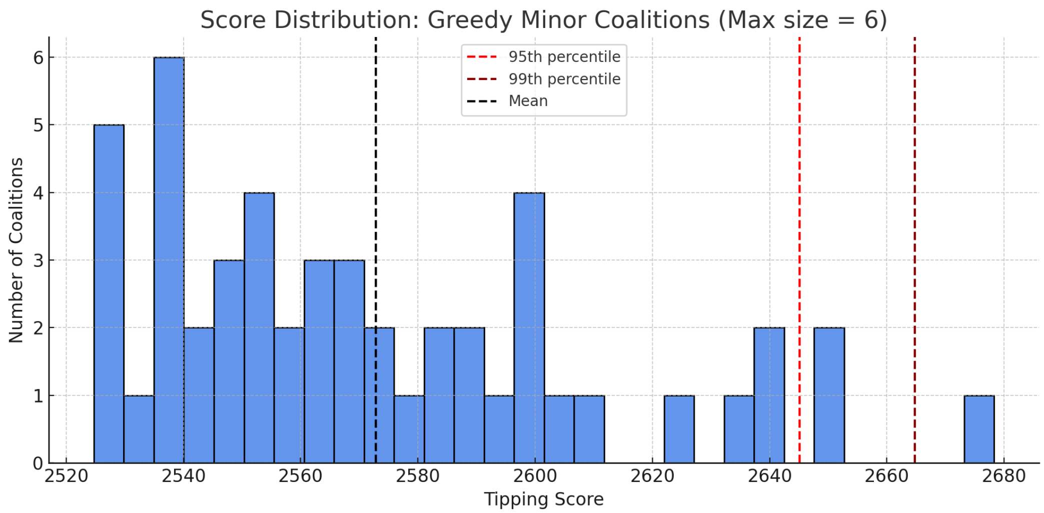

Figure 9: Distribution of tipping scores for top minor-player coalitions discovered using greedy forward-selection.

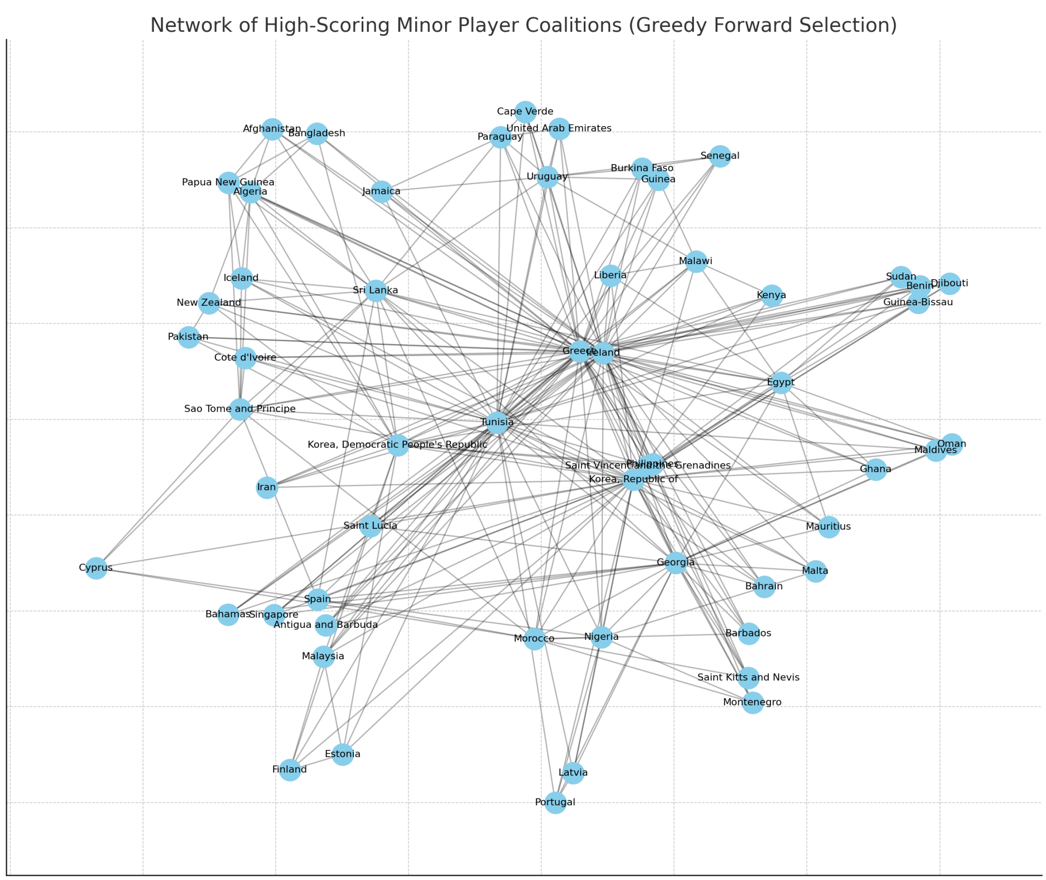

Figure 10: Network of minor players frequently co-occurring in high-scoring coalitions under the greedy search strategy.

- **Fossil Fuel Non-Proliferation Treaty (FF-NPT) Initiative** — a recent campaign led by vulnerable nations seeking to phase out fossil fuel production.

We found that several of our model-identified coalitions contain members from these influential real-world initiatives. For example, *Fiji*, *Tuvalu*, *Papua New Guinea*, and *The Bahamas* — all prominent AOSIS or FF-NPT members — appear frequently in high-scoring coalitions (see Table 1).

Interpretation. This overlap supports the model's validity: coalitions flagged as potentially influential by our tipping score framework are not only plausible but often reflect known diplomatic groupings that have, in fact, shaped global policy outcomes. The predictive alignment between our method and historical reality reinforces the model's interpretive and forecasting utility.

Limitations and Next Steps. Our matching focused only on membership overlap. Future work could extend this by analysing whether these coalitions took joint action within specific negotiation rounds, and whether their emergence preceded shifts in the positions of larger actors.

Initiative	Coalition Members	Tipping Score
High Ambition Coalition	Georgia, Maldives, Korea, Republic of, Ireland,...	2588.970124
High Ambition Coalition	Norway, Bulgaria, Korea, Republic of, Sri Lanka...	2479.820719
High Ambition Coalition	Maldives, Afghanistan, Papua New Guinea, Algeri...	2470.885138
High Ambition Coalition	Chile, Pakistan, Paraguay, Tunisia, Ireland, Gr...	2456.770215
AOSIS	Mauritius, Korea, Republic of, Egypt, Greece, I...	2601.572906
AOSIS	Sri Lanka, Papua New Guinea, Korea, Democratic ...	2601.343474
AOSIS	Papua New Guinea, Sri Lanka, Korea, Democratic ...	2601.343474
AOSIS	Algeria, Sri Lanka, Papua New Guinea, Korea, De...	2601.343474
AOSIS	Georgia, Maldives, Korea, Republic of, Ireland,...	2588.970124
Fossil Fuel Treaty	Pakistan, New Zealand, Korea, Democratic People...	2572.037733
Fossil Fuel Treaty	Solomon Islands, Papua New Guinea, Sri Lanka, A...	2493.596324
Fossil Fuel Treaty	Swaziland, Togo, Samoa, Tunisia, Ireland, Greece	2476.830569
Fossil Fuel Treaty	Jordan, Sudan, Samoa, Tunisia, Ireland, Greece	2471.643165
Fossil Fuel Treaty	Guinea-Bissau, Togo, Samoa, Ireland, Tunisia, G...	2467.402047

Table 1: Sample of high-scoring minor-player coalitions with overlaps to known real-world initiatives. Full results available at: `matched_real_world_coalitions.csv (GitHub)`

6.5 Reward Strategy Sensitivity Analysis

To evaluate the impact of reward design on reinforcement learning behaviour, we ran a sweep over key parameters in the tipping coalition environment. Specifically, we varied:

- **Size Cap**: The maximum allowed number of members in a coalition (4, 6, or 8);
- **Penalty** α: A reward penalty based on coalition size ($\alpha \in \{0.0, 1.0\}$);
- **Reward Threshold**: A tipping score cut-off (none, 500, or 800).

Each configuration trained a new policy from scratch using real data from the IEA ratification dataset and our tipping score function.

Findings. As shown in Figures 11 and 12, these strategy parameters strongly influence the learned coalition profiles:

- Without penalties, larger coalitions are favoured and reward values increase;
- Penalties incentivise smaller, more efficient groups with high per-member impact;
- Thresholds eliminate weak coalitions entirely, increasing selectivity.

Interpretation. This sensitivity analysis demonstrates that reward shaping is a crucial lever in tuning the realism and diversity of RL-discovered coalitions. Penalty and threshold constraints are essential for producing compact, strategic alliances that reflect the practical tradeoffs in real-world negotiations.

Figure 11: Mean tipping reward across RL experiments grouped by coalition size cap and size penalty.

7 Discussion and Implications

Interpret results, potential real-world applications, and links to climate governance frameworks.

8 Future Work

An important direction for future research is the principled determination of the tipping threshold λ_{crit}. Rather than specifying this value exogenously, one could calibrate it against historical data,

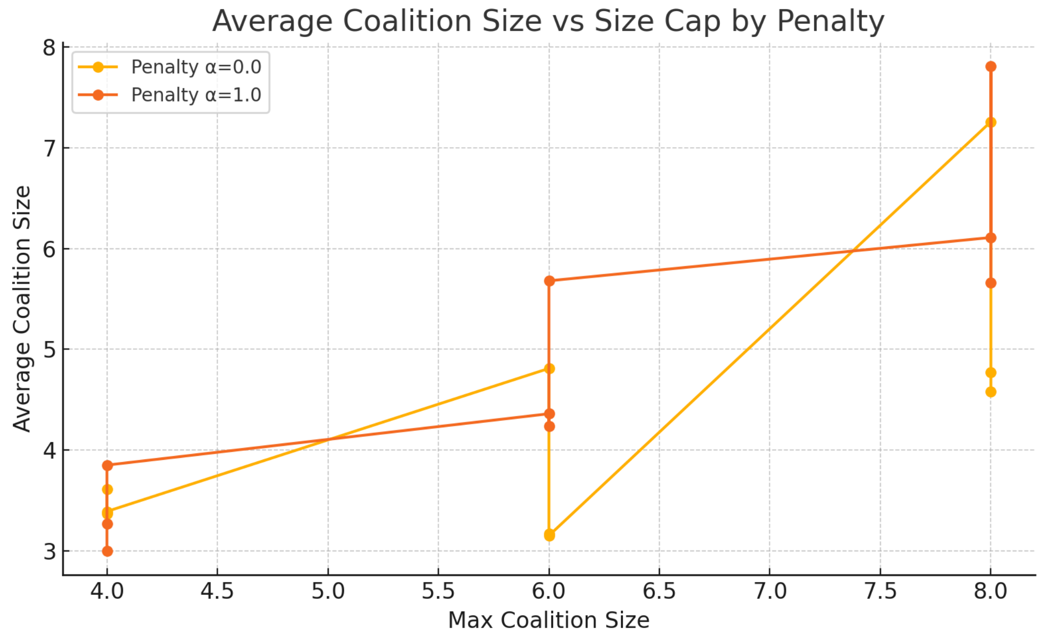

Figure 12: Average coalition size learned under different RL reward configurations. Penalty terms and caps steer the model toward more interpretable, compact coalitions.

Label	SizeCap	PenaltyAlpha	Threshold	MeanReward	AvgSize	NumCoalitions
cap=4_penalty=0.0_thresh=none	4	0.0	0	673.28	3.61	166
cap=4_penalty=0.0_thresh=500	4	0.0	500	976.79	3.37	155
cap=4_penalty=0.0_thresh=800	4	0.0	800	921.70	3.39	135
cap=4_penalty=1.0_thresh=none	4	1.0	0	556.57	3.00	166
cap=4_penalty=1.0_thresh=500	4	1.0	500	696.66	3.27	152
cap=4_penalty=1.0_thresh=800	4	1.0	800	890.55	3.85	133
cap=6_penalty=0.0_thresh=none	6	0.0	0	821.69	4.81	177
cap=6_penalty=0.0_thresh=500	6	0.0	500	961.02	3.17	141
cap=6_penalty=0.0_thresh=800	6	0.0	800	452.93	3.15	145
cap=6_penalty=1.0_thresh=none	6	1.0	0	618.04	4.36	172
cap=6_penalty=1.0_thresh=500	6	1.0	500	511.05	4.24	139
cap=6_penalty=1.0_thresh=800	6	1.0	800	462.88	5.68	140
cap=8_penalty=0.0_thresh=none	8	0.0	0	705.09	7.26	126
cap=8_penalty=0.0_thresh=500	8	0.0	500	688.36	4.77	108
cap=8_penalty=0.0_thresh=800	8	0.0	800	742.54	4.58	113
cap=8_penalty=1.0_thresh=none	8	1.0	0	513.93	6.11	192
cap=8_penalty=1.0_thresh=500	8	1.0	500	695.39	7.81	155
cap=8_penalty=1.0_thresh=800	8	1.0	800	997.87	5.66	133

Table 2: Summary of RL reward strategy experiments (mean values over 500 episodes).

derive it from the eigenvalue spectrum of baseline systems, or define it via bifurcation analysis of systemic behaviour. Such methods would ground coalition tipping detection in empirical or dynamical benchmarks, increasing interpretability and robustness.

9 Conclusion

Summarise findings, limitations, and next steps.

Acknowledgments

Optional — thank collaborators, code reviewers, etc.

Appendix: Final Coalitions Learned by Reinforcement Learning

The table below summarizes the final coalitions discovered by the reinforcement learning agent trained under different reward models. These were obtained after 1000 training epochs using a REINFORCE policy gradient method.

Table 3: RL-learned coalitions by reward strategy

Reward Function	**Final Coalition**	**Size**
Raw Reward	[0, 1, 2, 3, 4, 5, 6, 7, 8, 9]	10
Penalized Reward	[1, 2, 3, 4, 6]	5
Threshold-Based Reward	[1, 3, 5, 8]	4

These outcomes reflect the effect of reward shaping on learned policy structure. Threshold-based rewards reliably result in small tipping coalitions. Penalized rewards can balance efficiency and size when tuned appropriately, while raw rewards predictably encourage maximal inclusion.

Appendix: Cluster Composition and Internal Similarity

To support the analysis of country groupings derived from co-ratification patterns, we include visualisations for each of the eight identified clusters. These plots illustrate the internal diversity and cohesion of each cluster across three different perspectives:

- **PCA projections** show the relative similarity of countries within the cluster in two-dimensional space;
- **Dendrograms** expose the hierarchical relationships among members and suggest potential sub-clusters;

- **Cosine similarity heatmaps** provide a direct comparison of ratification overlap within the group.

Each visualisation is labelled using the compact descriptive cluster names introduced in Section 6. Cluster 5, which contains a notably large number of countries, demonstrates internal variation suggestive of three subgroups. These could be resolved in future work using finer-grained clustering methods or application-specific criteria.

The full country-to-cluster assignment table, including descriptive cluster labelling, is available at the project's GitHub repository:

`table_countries_by_cluster.csv (GitHub)`

This file contains all participating countries in the ratification dataset and their associated cluster classification, enabling cross-reference with any external indicators (e.g., emissions, population, development status).

Full cluster figures (PCA, dendrogram, heatmap) are provided in the supplementary materials folder: `figures/clusters/`.

Appendix: Greedy Algorithm Validation

To assess the quality of coalitions identified using greedy forward-selection, we compared the results to a complete brute-force reference dataset of all 2–4 member coalitions. Coalition scores were matched using normalised, sorted tuples of country names.

As shown in the summary and comparison plot (Figure 13), we found high consistency between greedy-selected coalitions and their ground-truth scores.

Interpretation. These results confirm that the greedy search method is both scalable and accurate for discovering tipping coalitions in large datasets. While it may occasionally overshoot the true impact of a coalition, its ability to identify high-scoring candidates without full enumeration makes it an invaluable tool for real-world strategic analysis Leskovec et al. (2014).

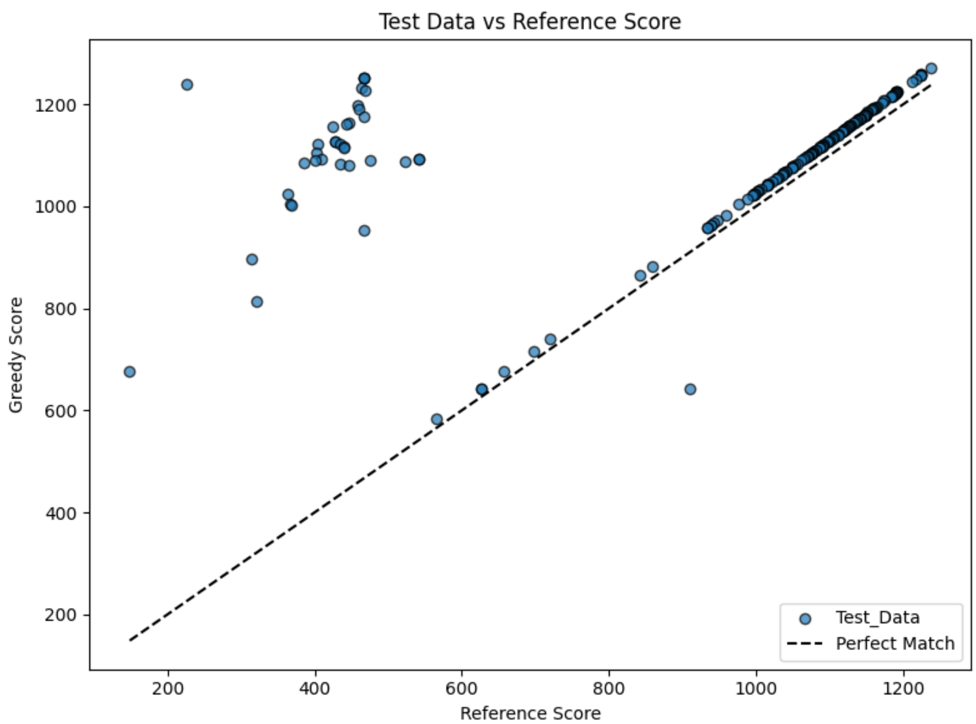

Figure 13: Comparison of tipping scores for coalitions discovered by greedy search versus full brute-force values. Most coalitions fall near the diagonal, indicating good agreement. A small number of outliers reflect local optima selected by the greedy algorithm.

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